

Reverse Engineering of Industrial Protocols From Network Traffic

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Abstract—Reliable protocol knowledge is often difficult to obtain in industrial networks, as industrial communications come with limited documentation, vendor-specific encodings, and opaque payloads. This lack of transparency hinders message interpretation and protocol analysis. To recover this missing protocol knowledge, network-trace-based protocol reverse engineering (PRE) infers message structure, field roles, and interaction logic directly from recorded traces. This enables protocol-aware intrusion detection, process monitoring, and protocol testing and fuzzing without access to device internals. Although PRE has advanced rapidly, existing techniques are developed under diverse objectives and assumptions. As a result, it is often unclear how isolated results relate to an end-to-end reverse-engineering workflow, and how evaluation outcomes should be compared across tasks and protocols. In this article, we cast reverse engineering of industrial protocols from network traces as a task-driven pipeline and articulate a unified task decomposition spanning message type identification, protocol syntax and semantic inference, payload pattern recognition and semantic inference, and protocol state machine reconstruction. For each task, we describe key methodological themes, common evaluation practices, and practical limitations that affect robustness and deployability in industrial settings. We further discuss security, privacy, and ethical risks that accompany increasingly capable PRE, and identify promising research directions toward more systematic, dependable, and deployment-oriented PRE methodologies.

Index Terms—Industrial communication protocols, industrial control systems (ICSs), network traffic analysis, protocol reverse engineering (PRE).

I. INTRODUCTION

INDUSTRIAL communication protocols constitute the backbone of modern industrial control systems (ICSs), enabling data acquisition, supervisory control, and coordinated operation

among heterogeneous industrial devices [1]. Despite their ubiquity in operational technology networks, many industrial deployments rely on proprietary, undocumented, or poorly specified protocols, which significantly hinders the reliable identification of message types, interpretation of communication content, and modeling of device interaction behaviors [2]. In particular, industrial message payloads encapsulate critical information, such as device states, measurement values, and control commands, yet their internal formats are often opaque due to proprietary implementations and vendor-specific design choices [3], [4]. This lack of protocol visibility fundamentally limits a principled understanding of system behavior, thereby constraining the effectiveness of essential ICS tasks, including system monitoring, security analysis, and fault diagnosis in real-world deployments [5], [6], [7].

Against this background, protocol reverse engineering (PRE) has emerged as a primary methodological paradigm for recovering protocol structure and communication semantics from observable artifacts. The existing PRE research can be broadly divided into network-trace-based and program-analysis-based approaches [8], [9]. Network-trace-based PRE focuses on inferring protocol structure and communication semantics solely from message exchanges observed on the network [10], [11], [12]. In contrast, program-analysis-based methods analyze internal artifacts, such as firmware images, binaries, or source code, to understand protocol implementations [13], [14], [15]. While powerful in controlled settings, program-analysis-based techniques are often impractical in industrial environments, where device internals are inaccessible, modifications are infeasible, and proprietary implementations provide limited structural transparency [16], [17]. Consequently, this article concentrates on network-trace-based PRE techniques, which better align with the operational constraints and closed-box nature of real-world ICS deployments.

By reconstructing protocol knowledge, PRE enables a wide range of downstream ICS applications. Recovered protocol structures facilitate automated testing and conformance verification through the generation of valid and invalid message sequences [18]. Building on reconstructed protocol state machines, PRE further enables stateful and semantic-aware fuzzing, significantly improving the effectiveness of vulnerability discovery [19], [20], [21]. In the security domain, PRE-derived semantics enhance intrusion detection by exposing protocol-violating behaviors, such as illegal function codes or parameter values, that statistic-based IDS cannot capture [22]. At the operational level, PRE supports system monitoring and process anomaly detection by revealing device states and control intentions directly from network traffic [23].

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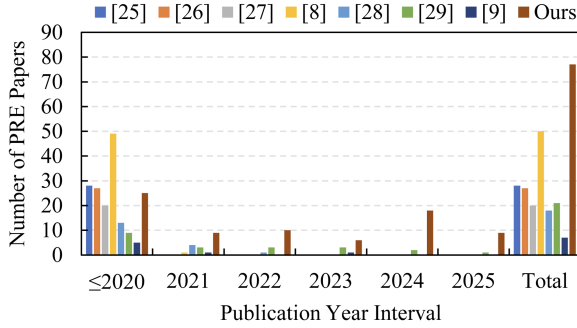


Fig. 1. Temporal distribution of PRE papers, illustrating the rapid growth of network-trace-based PRE in recent years and the limited coverage of this trend in existing surveys. Bars show the number of PRE papers reviewed across time periods by different survey works.

The existing surveys provide valuable foundations, yet they do not fully capture the rapid methodological advances made over the past decade. As shown in Fig. 1, network-trace-based PRE techniques for industrial protocols have expanded rapidly in recent years. Most of this progress has occurred in the past five years, yet it remains insufficiently reflected in prior surveys [8], [24], [25], [26]. More recent studies also exhibit limited scope. For example, work on controller area network (CAN) reverse engineering [27] offers detailed analysis but is confined to a single bus protocol and does not generalize to broader ICS settings. Research on industrial control PRE [28] primarily emphasizes earlier methods and provides only partial coverage of recent advances. Other efforts focus on automation aspects of PRE, such as interaction and information requirements [9], rather than on the technical foundations of PRE itself.

In contrast to prior work, this survey provides a contemporary and comprehensive view of PRE for industrial communication protocols. We base our study on papers published between 2016 and 2025 in leading security venues (S&P, CCS, USENIX Security, NDSS, TDSC, and TIFS) and major networking venues (SIGCOMM, MobiCom, INFOCOM, NSDI, JSAC, TMC, and TON), ensuring that our analysis captures the significant methodological progress made in recent years. Building on these recent advances, this article offers four major contributions. First, we present the first task-oriented taxonomy of PRE techniques, aligning algorithmic developments with practical reverse-engineering workflows and clarifying their roles in specific PRE tasks. Unlike prior surveys that organize techniques primarily by methodological categories, our taxonomy is structured around the PRE workflow and its constituent tasks. Second, we systematically describe techniques and evaluation practices across different PRE tasks and identify key open challenges. Third, we provide the first structured discussion of security, privacy, and ethical considerations of PRE in industrial settings. Finally, we outline concrete future research directions to support the continued development of PRE for ICSs.

The rest of this article is organized as shown in Fig. 2. Section II provides background on industrial protocols and introduces the PRE taxonomy used in this work. Section III reviews existing PRE techniques from a task-oriented perspective. Section IV reviews datasets used in prior industrial PRE research. Section V discusses security, privacy, and ethical considerations related to PRE in ICS settings. Section VI explores promising and emerging future research directions. Finally, Section VII concludes this article.

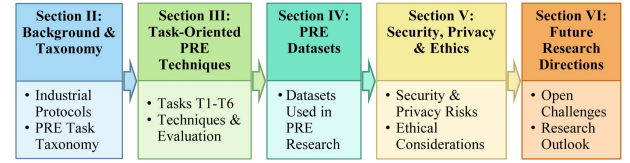


Fig. 2. Overview of the article organization.

TABLE I
KEY DIFFERENCES BETWEEN GENERAL-PURPOSE AND INDUSTRIAL COMMUNICATION PROTOCOLS FROM A PRE PERSPECTIVE

Dimension	General Protocols	Industrial Protocols
Specification openness	Open, standardized, and well-documented	Often proprietary, vendor-specific, or undocumented
Encoding format	Often self-describing formats (e.g., text-based or tag-based)	Compact binary; typically non-self-describing
Field granularity	Primarily byte-aligned fields	Combination of byte- and bit-level fields
Protocol structure	Often stateless or lightly stateful; more uniform message schema	Often stateful; formats are function- and context-dependent
Instruction / data model	General-purpose operations with relatively uniform data models	Rich command sets with heterogeneous and often nested data layouts
Payload size and fragmentation	Moderate payload sizes; fragmentation usually handled by underlying transport	Potentially large payloads; frequent application-layer multi-frame transfers
Traffic characteristics	More diverse traffic patterns	High volume but low diversity
Deployment variability	More uniform implementations due to mature standardization	Implementations vary by vendor, device type, and deployment configuration

II. BACKGROUND AND TAXONOMY

Although industrial communication protocols are routinely observed through network traces, the information required to understand their structure, semantics, and behavior is rarely explicit in the traffic itself. This mismatch between observability and inference underlies the challenges of PRE in industrial settings. This section examines the sources of this mismatch and introduces a task-oriented PRE taxonomy to organize the existing techniques.

A. Industrial Communication Protocols

Industrial communication protocols enable interactions among controllers, sensors, actuators, and supervisory systems in automated environments, such as production lines and process plants [29]. They span multiple layers of the communication stack. At the lower layers, fieldbuses and real-time industrial Ethernet systems, such as PROFIBUS, EtherCAT, and CAN, primarily carry cyclic I/O exchanges and time-critical process variables [27]. At the upper layers, industrial application protocols, including Modbus, OPC UA, DNP3, and S7comm, provide device semantics, configuration access, and supervisory information [28]. Together, these protocols constitute a heterogeneous industrial communication landscape shaped by control requirements, device diversity, and long-standing industrial technology stacks.

These characteristics fundamentally distinguish industrial protocols from general-purpose IT protocols. As summarized in Table I, these distinctions manifest at multiple levels. At the representation level, industrial protocols commonly rely on proprietary specifications and compact, nonself-describing binary

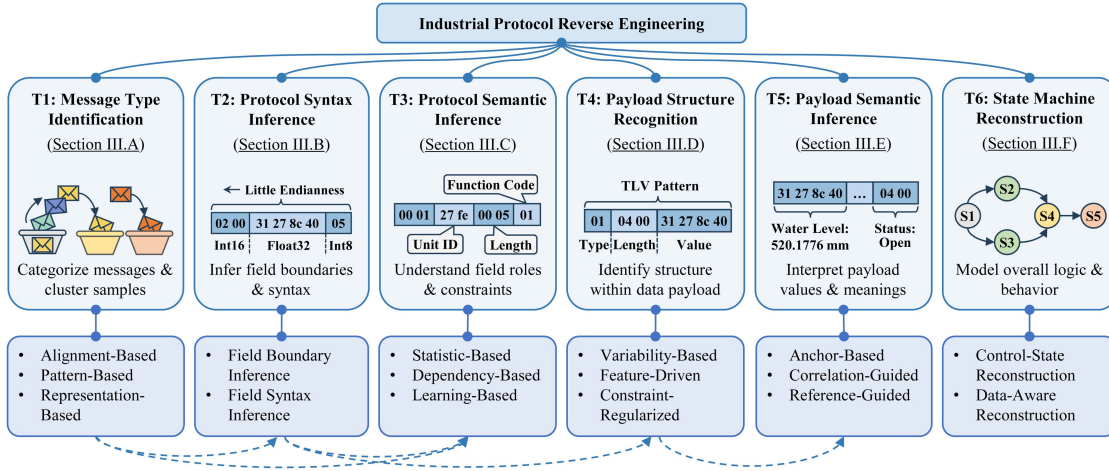


Fig. 3. Task-oriented taxonomy of industrial PRE, illustrating the major inference tasks in the PRE pipeline for recovering protocol structure, semantics, and behavior. Each task is characterized by representative technique families; the dashed connections indicate cross-task integration.

encodings that mix byte- and bit-level fields [6], [7]. At the structural level, message formats are often context-dependent, varying with protocol functions and service states, and support heterogeneous command sets and data layouts [30]. At the traffic level, industrial communications are dominated by cyclic control exchanges, resulting in high-volume but low-diversity patterns, while process variable exchange and diagnostic operations may involve larger or segmented payloads [31]. Beyond protocol design, implementations further vary across vendors, device types, and deployment configurations [16], [19].

These properties make industrial communication protocols difficult to interpret directly from network traces. Understanding protocol operation in practice therefore cannot rely on explicit specifications or self-describing message formats. Instead, it requires inference-based methods that reconstruct message formats, functional roles, and communication patterns from limited observational evidence. Under these conditions, PRE becomes essential for obtaining reliable insight into ICSs.

B. Taxonomy of Industrial PRE

In this article, PRE refers to network-trace-based PRE, which aims to recover the structure and semantics of industrial communication protocols directly from observed network traffic, without relying on official specifications or source code [25]. In industrial settings, PRE operates on raw packet streams collected from operational networks and seeks to infer how messages are differentiated, how protocol fields are organized, and how exchanged data corresponds to device behavior or underlying process operation. Due to the limited observability inherent in network traces, such inference should be performed under incomplete and implicit protocol information, motivating a systematic decomposition of PRE into well-defined inference tasks.

To structure such inference, we organize industrial PRE into a task-oriented taxonomy, as illustrated in Fig. 3. The taxonomy follows the logical progression of a typical PRE pipeline, in which protocol knowledge is incrementally recovered from network traces [10], [25], [32]. It starts with message type identification to separate heterogeneous traffic, providing a coarse-grained organization of observed messages. Building on this separation, protocol syntax and semantic inference focus

on recovering message layouts and associating protocol fields with their functional roles. The analysis then moves to the payload level, where payload structure and semantic inference extract finer-grained data representations and interpret payload contents in relation to device states or process variables. Finally, protocol state machine reconstruction captures temporal and logical dependencies across message sequences, modeling the overall protocol behavior. This task-oriented organization clarifies dependencies among different inference problems and provides a systematic framework for reviewing existing PRE techniques in the following section.

III. TASK-ORIENTED PRE TECHNIQUES

This section reviews industrial PRE techniques from a task-oriented perspective. Rather than attempting to recover complete protocol specifications in a single step, existing PRE systems typically address concrete reverse-engineering objectives through a sequence of interdependent tasks [6], [33], [34]. Accordingly, we organize prior work by the specific PRE tasks they target. For each task, we summarize representative techniques, commonly used evaluation metrics, typical performance analysis practices, and key open challenges. We do not present fine-grained quantitative comparisons across methods, as existing studies are evaluated on diverse protocols and datasets whose scale and traffic quality are difficult to normalize. Moreover, there is currently no widely accepted framework for assessing PRE data quality [27], [35].

A. Message Type Identification

Message type identification aims to partition protocol messages into homogeneous groups, where each group corresponds to a distinct message type with a consistent structural or functional role. This task serves as the entry point of most network-trace-based PRE pipelines, which assume messages of different types have been separated.

1) *Techniques*: In practice, message type identification is commonly formulated as an unsupervised clustering problem under the constraint that no protocol specification is available. Existing approaches mainly differ in how message similarity is constructed and which aspects of the message are emphasized

to mitigate payload-induced variability. Based on these design choices, prior work can be broadly categorized into alignment-based, pattern-based, and representation-based approaches.

Alignment-based methods: Alignment-based methods identify message types by comparing messages positionally using sequence alignment or format comparison, under the assumption that messages of the same type share similar byte values at corresponding offsets [36]. Message similarity is typically computed via global byte-wise alignment, and clustering is then performed based on the resulting pairwise similarity matrix, most commonly using hierarchical clustering [37]. Several studies refine raw byte-wise alignment by introducing segment-aware comparison [38] or improved alignment spaces to better accommodate binary and industrial protocols [39]. While these methods can capture fine-grained structural similarities, they incur high computational overhead due to exhaustive pairwise comparisons [37]. They are also particularly sensitive to payload-induced distortion when variable-length or data-heavy fields dominate message contents [38].

Pattern-based methods: Pattern-based methods relax strict positional alignment by constructing message similarity from statistical or structural patterns [40]. A representative line of work focuses on probabilistic keyword inference. It models messages using discriminative byte patterns whose occurrence statistics distinguish different message types, and derives similarity from shared keyword distributions [34]. In industrial protocol settings, some approaches exploit coarse-grained structural regularities, such as message length distributions or stable control regions, to reduce payload-induced interference [41]. More recent work automates this process by identifying entropy-stable header regions via change-point detection and performing clustering on the extracted headers, improving robustness under highly dynamic payloads [42], [43]. By relying on stable patterns rather than global positional consistency, pattern-based methods exhibit improved resilience to payload variability, while implicitly assuming that such patterns are sufficiently discriminative [34], [43].

Representation-based methods: Representation-based methods adopt a data-driven perspective by learning latent representations of protocol messages through neural models, rather than relying on manually designed similarity measures [44]. Autoencoders and shared-learning frameworks are employed to embed messages into latent spaces that capture structural regularities across message instances [45]. Subsequent studies leverage transformer-based architectures to learn contextualized message embeddings, and perform message type identification via clustering in the learned representation space [46]. Dynamic and hierarchical inference frameworks further integrate learned representations with iterative refinement mechanisms to improve type separation under complex protocol behaviors [5], [47]. Although learned representations can capture complex and nonlinear similarities, their effectiveness depends on the availability of sufficient training data, and their interpretability remains limited [44], [46].

2) Evaluation Metrics and Performance: Message type identification is commonly evaluated as a clustering task by measuring the consistency between inferred clusters and ground-truth message types. When ground-truth labels are available, studies typically report metrics, such as homogeneity, completeness, and their harmonic mean (V-measure) [43], as well as precision, recall, or F1-score [45]. In the settings without labels, adjusted Rand index or intrinsic clustering metrics such as silhouette score are often adopted [46]. Overall, the existing methods

achieve strong performance on well-structured protocols, particularly when message headers are stable and payload variability is limited. However, performance tends to degrade for protocols with highly dynamic payloads, weakly defined headers, or large numbers of message types, indicating that message type identification remains challenging in complex industrial and proprietary environments.

3) Challenges: Despite steady progress, message type identification remains challenged by constructing robust similarity measures in the presence of highly dynamic payloads, scalability issues arising from pairwise similarity computation or model training, and limited generalization across diverse protocol families. Addressing these challenges is essential for reliable PRE in realistic industrial environments.

B. Protocol Syntax Inference

Protocol syntax inference aims to recover the internal structural organization of protocol messages by identifying how each message type is partitioned into individual fields and how these fields are interpreted. It operates at the intra-message level, focusing on field layouts and decoding rules, and provides the foundation for subsequent semantic inference.

1) Techniques: Protocol syntax inference typically involves two aspects: field boundary inference and field syntax inference. Field boundary inference identifies where fields begin and end in a message, including each field's position, length, and granularity. Field syntax inference then determines how the extracted fields should be interpreted, such as their data type, encoding, and byte order. Prior studies differ in which aspects they address and what evidence they exploit for structural reconstruction.

Field boundary inference: The early field boundary inference approaches are largely driven by cross-message alignment. Multiple messages of the same type are aligned, and conserved regions are interpreted as fields, with boundaries inferred from alignment gaps or divergence points [48]. Bit-oriented extensions subsequently adapt alignment-based inference to finer granularities by redefining matching and backtracking rules to support subbyte fields [49]. This alignment-based paradigm is further extended to wireless and physical-layer traces by aligning messages via inferred preamble or synchronization patterns, enabling boundary recovery beyond byte-aligned settings [50].

To reduce reliance on cross-message alignment, a substantial body of work infers field boundaries by analyzing statistical regularities within individual messages. NEMESYS identifies candidate boundaries by exploiting intrinsic value-change patterns and repetition structures [51], while later refinements employ variance correlation and principal component analysis to iteratively adjust coarse segmentations [52]. Other approaches formulate boundary detection as a probabilistic inference problem, using entropy, mutual information, or expert voting mechanisms to identify likely boundary positions [53], [54], [55]. Domain-specific adaptations further exploit structural characteristics of industrial protocols, such as compact headers and short control fields, to guide iterative merging or splitting of candidate segments [56]. More recent work frames boundary inference as a sequence labeling or segmentation problem and applies deep learning models to identify byte- or bit-level boundaries [12], [57]. These models can improve accuracy on complex binary protocols, but typically require more data and supervision.

Field syntax inference: Compared to field boundary inference, fewer studies explicitly target field syntax inference. Early efforts incorporate semantic information and contextual

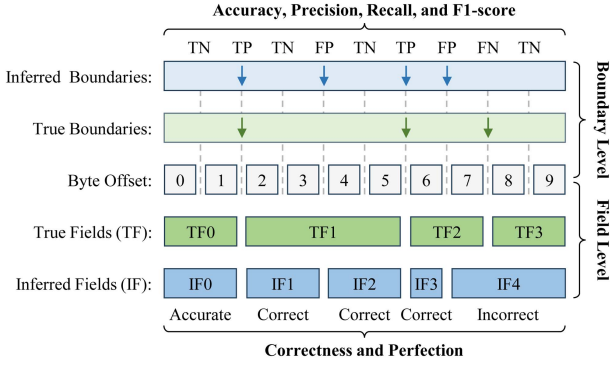


Fig. 4. Boundary-level and field-level evaluation in protocol syntax inference. Boundary-level evaluation treats each candidate position as a binary decision, while field-level evaluation matches inferred fields to ground-truth fields.

constraints to infer field interpretation rules alongside boundary detection, enabling limited recovery of data types and field roles [58]. Subsequent work explores model-based and knowledge-driven techniques for inferring field grammar, leveraging protocol templates, expert knowledge, or probabilistic constraints to guide interpretation [11], [59]. More recently, graph-based and relational reasoning methods represent inferred fields and their relationships as structured graphs, enabling syntax inference via representation learning over field dependencies [30], [60]. Overall, these approaches demonstrate that PRE can move beyond boundary recovery toward richer syntax reconstruction, but they often depend on auxiliary knowledge or domain-specific assumptions.

2) Evaluation Metrics and Performance: Protocol syntax inference is evaluated at different granularities, which leads to distinct evaluation formulations, as illustrated in Fig. 4. At the *boundary level*, each candidate position between adjacent bytes (or bits) is treated as a binary decision (boundary versus nonboundary). True positives, false positives, true negatives, and false negatives are therefore defined over boundary positions, and metrics, such as accuracy, precision, recall, and F1-score, are computed accordingly [12]. At the *field level*, evaluation is instead based on matching inferred fields to ground-truth fields. NetPlier [34] defines an inferred field as *correct* if it is part of a single true field or merges several consecutive true fields, and as *accurate* if it perfectly matches a true field. Based on these notions, it reports *correctness* as the fraction of correct fields and *perfection* as the fraction of accurate fields among all true fields.

These two evaluation views capture different failure characteristics: boundary-level metrics primarily reflect local delimiter errors, whereas field-level metrics reflect end-to-end format quality. Accordingly, reported performance depends strongly on both the targeted inference granularity and the complexity of the evaluated protocols. Boundary inference methods often achieve high accuracy on protocols with stable formats, whereas methods that aim to recover complete field structures or syntax face substantially greater challenges on complex and variable binary protocols. As a result, performance figures vary considerably across studies, owing to differences in datasets, evaluation protocols, and the scope of syntax inference considered.

3) Challenges: A fundamental technical challenge in protocol syntax inference is the strong dependence of syntax recovery on boundary accuracy. Errors in field segmentation directly

propagate to syntax inference, making it difficult to reliably infer data types, encodings, and byte order when boundaries are imperfect. Moreover, the limited availability of field-level evaluation signals further complicates the validation of inferred syntax, hindering systematic progress beyond boundary-centric solutions.

C. Protocol Semantic Inference

Protocol semantic inference aims to determine the operational roles played by recovered fields in protocol execution, rather than only their structural boundaries or encodings. Typical targets include message type indicators, function codes, length-related fields, and transaction-related identifiers. For example, a 1-byte field that takes values, such as 0x01, 0x03, and 0x05 across different messages may be recognized structurally as a compact numeric field during syntax inference, but semantic inference further asks whether these values denote message categories, protocol operations, or transaction states.

1) Techniques: The existing protocol semantic inference approaches mainly differ in how protocol-level semantic roles are inferred from observed traffic. Accordingly, prior work can be broadly categorized by the type of evidence exploited for semantic inference, including statistic-based, dependency-based, and learning-based approaches.

Statistic-based semantic inference: Statistic-based approaches infer protocol semantics through explicit statistical associations between field values and observable communication properties. FieldHunter [33] is a representative example that assigns semantic roles by correlating candidate fields with message length, endpoint identity, and request-response structure. Based on measures of variability and correlation, it identifies roles, such as message type indicators, length fields, identifiers, transaction-related fields, and accumulators. Follow-up work shows that similar statistic-based reasoning can still support coarse-grained semantic inference even when protocol visibility is limited, such as in partially encrypted or obscured traffic [61].

Dependency-based semantic inference: Dependency-based approaches infer protocol semantics by analyzing how fields function within interactions. Early work shows that semantic roles can be associated with fields by examining their involvement in request-response relations and conditional message structures [62]. The two-pathway model [63] further generalizes this idea by jointly considering field stability and interfield dependencies. More recent work emphasizes control or discriminator fields and shows that a small subset of fields governs protocol behavior. Specifically, control-field-driven inference exploits interaction dependencies to associate fields with roles, such as function codes or operation selectors [6]. **Learning-based semantic inference:** As illustrated in Fig. 5, learning-based approaches infer protocol semantics by learning mappings between data patterns and semantic roles from existing protocols and transferring this knowledge to previously unseen protocols. Prior work shows that protocol-level field roles can be inferred by grouping fields with similar learned representations [64]. FSIBP [65] and PREIUD [66] further extend this direction by leveraging feature learning and representation learning to automate field-role inference for binary and industrial protocols. Complementary approaches exploit recurring patterns and temporal regularities to learn semantic signatures of fields and associate them with protocol roles in an inductive manner [7], [67]. Empirical studies on real-world protocols further confirm

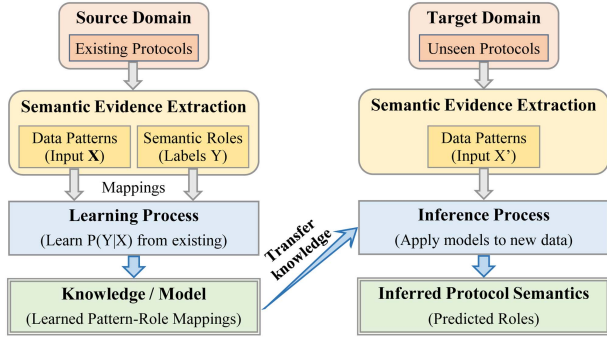


Fig. 5. Learning-based protocol semantic inference via cross-protocol knowledge transfer, where semantic mappings learned from existing protocols are transferred to infer semantic roles in unseen protocols.

that such learning-based semantic inference can generalize to protocols without available specifications [68].

2) **Evaluation Metrics and Performance:** Protocol semantic inference lacks standardized evaluation methodologies. Existing studies typically adopt problem-specific evaluation criteria, including agreement with known protocol specifications, or qualitative validation through manual analysis. When ground-truth semantic annotations are unavailable, inferred semantics are often evaluated indirectly by assessing their utility in downstream tasks, such as protocol-aware intrusion detection, fuzzing, or traffic parsing [21], [69], [70]. Consequently, reported performance varies widely across studies and is difficult to compare in a unified manner, as results are strongly influenced by protocol complexity, traffic diversity, and the semantic granularity considered. Existing approaches are generally more stable for recurring semantic roles, such as length fields, sequence numbers, or control codes. Protocol-specific semantics, such as the ROSCTR and Function Group fields in S7comm, are harder to infer reliably because reference instances are limited.

3) **Challenges:** A key challenge in protocol semantic inference lies in the absence of a unified semantic taxonomy for protocol fields. Without commonly accepted semantic categories and reference definitions, inferred field semantics lack stable anchors, making it difficult to establish transferable semantic models from traffic. For example, message type indicators, operation selectors, and transaction identifiers may all fall under control-related semantics, yet their semantic distinctions are often defined inconsistently across studies. This limitation is particularly severe for protocol-specific semantics, where scarce and nonreusable evidence in observed traces further constrains reliable semantic assignment.

D. Payload Structure Recognition

Payload structure recognition recovers recurring structural patterns within message payloads. It typically outputs bit/byte-level offset-length segments. For long or composite payloads, it can further recover repeated submessages and nested blocks under explicit length constraints. Compared with protocol-level syntax inference, it focuses on semantics-heavy payload regions that carry application data and complex parameters but expose few explicit delimiters. The recovered structure provides a basis for payload semantic inference.

1) **Techniques:** Existing payload structure recognition approaches aim to recover field boundaries by analyzing repeated

payload observations and enforcing consistent layouts across messages. While they share this common objective, existing methods differ primarily in the type of structural evidence used to infer boundaries and in how segmentation hypotheses are regularized. Based on these differences, prior work can be broadly grouped into three paradigms: variability-based segmentation, feature-driven recognition, and constraint-regularized recognition.

Variability-based segmentation: A common strategy infers payload boundaries from variability observed across repeated payload samples. It computes per-bit or per-byte variability statistics, such as flip rates, transition counts, and distributional stability, and then locates field boundaries by detecting abrupt changes of these statistics along the payload [71], [72]. Robust variants move beyond a single global profile and model each position as a short sequence across observations. They use interposition similarity and correlation to decide whether adjacent bits or bytes belong to the same field [73], [74]. These methods work well when traffic contains abundant repetitions, but their effectiveness can degrade when signal changes are rare in the capture window or layouts vary across contexts.

Feature-driven recognition: Recent approaches increasingly treat segmentation as a data-driven grouping problem. They encode each bit position, byte position, or small position block as a feature vector that captures multiple behavioral aspects, including variability, value distribution, and simple temporal descriptors [75]. Boundaries are then derived from transitions between learned groups or directly predicted from these representations [76], [77], [78]. Related ideas also appear in interactive closed-box settings, where response-driven signals provide weak supervision to localize influential payload snippets and approximate their boundaries across queries [69], [70], [79]. Compared with fixed heuristics, feature-driven methods better handle heterogeneous field types and unknown numbers of fields, but they shift sensitivity to feature design and data representativeness.

Constraint-regularized recognition: For long or composite payloads, structure is better revealed by explicit format constraints and cross-sample repetition than by variability-based cues alone. Constraint-driven methods identify control fields that govern block extents and validate inferred boundaries against observed payload spans. Along the same line, template-centric recognition aligns repeated records or submessages to expose stable anchors and variable regions, and iteratively refines boundaries across samples [80], [81]. Multiple-sequence alignment extends this idea by revealing conserved cores and insertions and deletions, enabling boundary discovery when explicit delimiters are absent [82].

2) **Evaluation Metrics and Performance:** When ground-truth boundaries are available, segmentation quality is commonly measured by boundary precision, recall, and F1-score at bit or byte granularity, often using a small tolerance window to account for minor alignment ambiguity [71], [72], [73]. For methods that output position groupings, consistency can be evaluated by the agreement between predicted and reference group labels [76]. For composite payloads, evaluation further checks whether inferred length-controlled extents match observed payload spans and whether recovered templates remain stable across samples [80], [81].

Practical performance is mainly shaped by sample efficiency, scalability with payload length, and robustness under mixed layouts. Variability-based methods typically require abundant

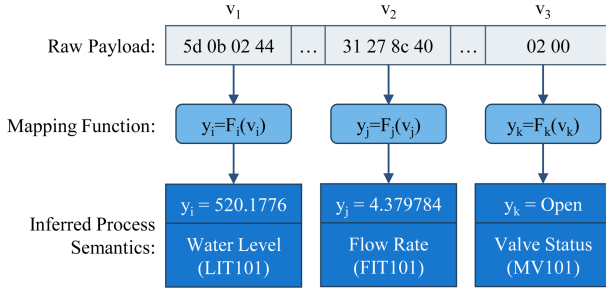


Fig. 6. Example of payload semantic inference. Raw payload data are analyzed to recover internal representations and to infer process semantics, where decoding functions associate payload content with interpretable values and domain meanings.

repeated observations to obtain reliable position-wise statistics [71], [72], and can degrade when traffic diversity is limited or layouts vary across contexts [74]. Feature-driven methods reduce the need for abundant repeated observations, but they incur additional costs for feature representation learning. Their performance is also sensitive to the choice and quality of features [76], [77], [78]. Constraint-regularized methods can stabilize long or repeated structures but introduce additional validation and alignment overhead [81], [82]. ReCAN supports reproducible comparisons of metrics and scalability across traces and setups [83].

3) Challenges: Mixed layouts created by optional fields and context dependent multiplexing blur structural cues. At a fixed offset, the same bits or bytes can correspond to different fields across messages. As a result, variability statistics and feature representations mix multiple structures. The resulting change signals are blurred rather than localized [74], [77]. This ambiguity also destabilizes cross message regularization. Alignment, template induction, and length controlled block validation must reconcile hypotheses that fit only subsets of samples. As a result, inferred boundaries become unstable and recovered blocks become inconsistent across observations [80], [81], [82]. Bit-level packing and endianness further weaken usable anchors and reduce contrast in variability profiles, which compounds the difficulty of locating reliable boundaries [71], [73].

E. Payload Semantic Inference

As illustrated in Fig. 6, payload semantic inference aims to assign domain meanings to recovered payload fields and to derive their decoding rules. Unlike protocol semantic inference that mainly targets control-oriented header fields, it focuses on semantics-heavy data regions grounded in physical processes and device behaviors.

1) Techniques: Existing payload semantic inference approaches aim to assign domain meanings to recovered payload fields and infer their decoding rules. They differ mainly in the type of evidence used to ground semantic interpretations. In practice, three main lines of work have emerged: anchor-based semantic alignment, correlation-guided semantic mining, and reference-guided semantic transfer.

Anchor-based semantic alignment: A practical way to obtain semantics is to use standardized diagnostic services or professional tooling as semantic anchors. The key idea is to trigger or observe queries whose returned values have known meanings, and then align these reference values with candidate

payload fields to infer both field roles and decoding rules. For in-vehicle diagnostics, DP-Reverser leverages professional diagnostic tools and automated physical triggering to recover message semantics and proprietary formulas for interpreting responses at scale [31]. ACTT and CAN-D similarly exploit diagnostic knowledge to attach physical meanings to a subset of extracted signals and to refine encoding details that are hard to infer from traffic alone [32], [84]. This family provides high-confidence semantic grounding, but its coverage is constrained by what standards expose and what tooling can exercise in the captured operational context.

Correlation-guided semantic mining: When explicit semantic anchors are unavailable, another line of work infers semantics by correlating payload fields with side information that reflects the underlying physical system. LibreCAN combines raw CAN traces with smartphone inertial measurements and OBD-II readings to match candidate signals to known physical quantities [85]. AutoCAN further exploits laws of physics by searching for correlation and functional relationships among time series, enabling label propagation once a small set of signals are identified [86]. For industrial control protocols, SeMiner uses image-based process observations to extract semantic channels and match them to payload fields, enabling domain semantics inference without protocol specifications [87]. Recent work also moves beyond correlation to directional reasoning. PicaCAN models physically induced causality and uses time-precedence structure to infer physical semantics of signals from passive traces [4]. In wireless cyber-physical control, similar correlation-based reasoning can be used to associate captured command payloads with control actions [88]. These approaches generalize across proprietary encodings, but they depend on the availability, synchronization quality, and state coverage of side information.

Reference-guided semantic transfer: It aligns unknown traffic with reference patterns in the form of previously decoded or annotated message templates from related systems. This establishes cross-system correspondences and enables the transfer of semantic labels and decoding rules. CANMatch exploits frame reuse and similarity across vehicles [89]. It infers signal meanings and scaling factors through automated frame matching and redundancy resolution. Translation-oriented systems, such as LibreCAN, can also benefit from this paradigm once a small set of semantically grounded signals is available [85]. This enables broader interpretation of raw traces via label propagation. While reference-guided transfer reduces the need for direct semantic anchors, it remains sensitive to deployment variability, partial reuse, and mismatches introduced by vendor-specific multiplexing.

2) Evaluation Metrics and Performance: Semantic inference is typically evaluated along three axes: correctness of semantic assignment, fidelity of decoded values, and coverage. When reference semantics are available through diagnostics, tooling, or instrumented observations, studies measure whether each inferred field matches the intended quantity using top-1 or top- k accuracy [32], [84]. They also compare decoded time series against references using MAE, RMSE, and correlation after applying inferred signedness, endianness, scaling, and units [85], [87]. Coverage is reported as the fraction of fields or signals that can be assigned semantics and decoded reliably under the available evidence [32], [86].

Anchor-based alignment tends to achieve higher correctness and value fidelity when anchors can be exercised, while coverage

is bounded by what standards or tools expose and what operational states can be triggered or observed during capture [32], [84]. Correlation-guided mining is more broadly applicable but is sensitive to synchronization quality, delays, noise, and limited state coverage, which can reduce fidelity or introduce false matches [85], [88]. Physics- and causality-guided designs can improve robustness over correlation alone when sufficient data support stable relationship or precedence estimation [4], [86]. Reference-guided semantic transfer can be accurate once cross-trace matches exist, yet its effective coverage and mapping stability depend on cross-system reuse and resilience to mode shifts [89].

3) Challenges: Payload semantic inference is fundamentally underconstrained because multiple fields can exhibit similar temporal patterns while encoding different quantities, and decoding rules such as scaling, offset, and signedness may be indistinguishable without semantic anchors [90]. Since passive traces reveal only value evolution rather than physical meaning [31], temporally similar fields may remain observationally equivalent, allowing multiple decoding hypotheses to explain the same trace data equally well. As a result, semantic assignment and decoding rule inference cannot be uniquely determined from network traffic alone without reliable anchors.

F. Protocol State Machine Reconstruction

Protocol state machine reconstruction aims to infer a compact behavioral model of a protocol from network traces, typically as a finite-state machine (FSM), or as an extended finite-state machine (EFSM), with variables and guards. An FSM captures control-state evolution and ordering constraints over abstracted message symbols, while an EFSM additionally models value-dependent branching through guarded transitions and variable updates.

1) Techniques: Most approaches first abstract raw messages into a finite symbol set and then learn a state machine over the resulting sequences. Prior work mainly falls into control-state reconstruction (FSM) and data-aware reconstruction (EFSM).

Control-state reconstruction: Control-state reconstruction learns an FSM that encodes which message classes can follow one another and how requests pair with responses. A typical workflow consists of message abstraction, state construction from symbol sequences, and state merging or splitting to obtain a compact model that generalizes beyond the observed network traces [91], [92], [93]. Network traces are often noisy and underconstrained because of partial state coverage, payload variability, and session interleaving. To cope with this, many systems embed robustness directly into the reconstruction loop. They adopt probabilistic formulations to tolerate noisy observations [94] or analyze how abstraction granularity and merging criteria affect model quality, helping suppress spurious states and transitions [95]. In addition, inferred control-state machines are validated and refined through downstream tasks, such as stateful testing or fuzzing, where the learned model guides sequence generation and exploration of deeper behaviors [16], [96].

Data-aware reconstruction: Data-aware approaches first learn a control-state skeleton from abstracted message sequences. They then lift this skeleton to an EFSM by selecting payload fields as variables and inferring transition guards and update rules that explain value-dependent branches across traces [97]. Because guard and update inference from passive traces is highly underconstrained, EFSM reconstruction often

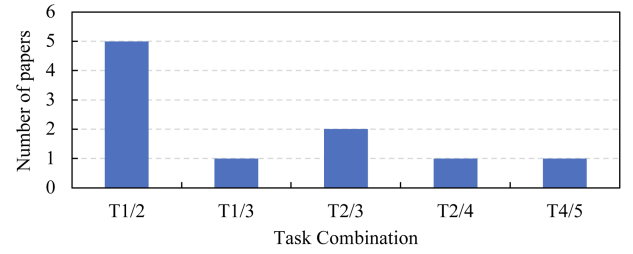


Fig. 7. Cross-task patterns observed in prior PRE research. Tx/y denotes the combination of tasks x and y.

relies on constraint-based refinement or guided correction to revise candidate predicates and state distinctions [98].

2) Evaluation Metrics and Performance: When a reference FSM or EFSM is available, studies typically evaluate inferred models using state precision and recall, transition precision and recall, completeness, F1-score, and model conciseness under a consistent abstraction [97]. In the absence of a reference model, evaluation commonly relies on held-out traces. Typical metrics include trace or session acceptance rate, next-symbol prediction accuracy or likelihood, and model size [92], [94].

Across both settings, performance is largely determined by the chosen abstraction and by how states are merged or split, reflecting a fundamental tradeoff between state explosion and state aliasing [95]. Probabilistic formulations can improve robustness to noise and unseen variations, but they may overgeneralize when trace diversity is limited [94]. EFSM reconstruction, by contrast, can better capture value-dependent behavior, yet it remains sensitive to ambiguity in guard and update inference from passive traces [97], [98].

3) Challenges: The core challenge in protocol state-machine reconstruction lies in the nonidentifiability of latent control structure from finite execution traces. Because traces expose only observable message sequences, distinct state machines can exhibit identical external behavior, making reconstruction sensitive to abstraction choices, such as state merging or splitting. This challenge is further amplified in data-aware reconstruction, where lifting an FSM skeleton to an EFSM introduces additional degrees of freedom. Different selections of variables and guard or update predicates can fit the same traces, yielding structurally different yet observationally equivalent models [97], [98]. Consequently, reconstruction outcomes are difficult to compare across studies [95].

G. Cross-Task Integration and Dependencies

To complement the per-task review above, Fig. 7 summarizes cross-task patterns in prior PRE studies. Among studies that span multiple tasks, T1/T2 is the most common combination, followed by T2/3, while other pairings, such as T1/3, T2/4, and T4/5, appear only sporadically. This pattern reflects the PRE pipeline's dependency structure: cross-task coupling arises when upstream tasks produce stable intermediate outputs for downstream analysis, such as message partitions, field boundaries, and structural hypotheses. By contrast, payload semantic inference and state machine reconstruction rely on heterogeneous evidence, including application-specific semantics, side information, and longer-range temporal context, for which no equally stable intermediate abstraction is available. Consequently, these later-stage tasks are often only partially grounded in upstream outputs and, in many cases, require additional

TABLE II
DATASETS USED IN INDUSTRIAL PRE RESEARCH

Dataset	Year	Protocols	Tasks	Scale
[65]	2024	7 protocols (e.g., DNP3, Modbus, S7comm, OMRON-FINS)	T2, T3	6,600 messages
[99]	2024	21 protocols (e.g., DNP3, Modbus, S7Comm, PMU)	T2, T3	19–82k+ packets per protocol
[10]	2023	10 protocols (e.g., DNP3, Modbus)	T2, T4	10,000 messages
[34]	2021	10 protocols (e.g., DNP3, Modbus)	T1, T2, T6	9,453 messages
[100]	2017	Water-treatment CPS (six-stage testbed; Ethernet/IP)	T5, T6	11 days; 946,722 samples
[101]	2017	Water-distribution CPS (three-stage testbed; Modbus)	T5, T6	16 days; 123 sensors/actuators

evidence rather than continuing earlier-stage inference. Later reasoning stages therefore inherit intermediate assumptions that are often difficult to verify or revise, which may affect downstream robustness.

IV. DATASETS FOR INDUSTRIAL PRE

Table II summarizes datasets used in prior industrial PRE studies. Viewed through the task-driven PRE pipeline, these datasets support different stages of protocol knowledge recovery. For message- and field-level tasks, datasets are often built from protocol traces whose annotations are obtained with the aid of protocol analyzers, such as Wireshark and manual verification. Such resources support message typing, syntax recovery, and field-level semantic analysis. By contrast, datasets, such as SWaT [100] and WADI [101], provide process-aware evidence, including network traffic, sensor/actuator readings, and operational logs collected over sustained plant execution, and are therefore more relevant to payload semantic inference and behavior-oriented analysis.

Public dataset support remains uneven across PRE tasks. Dataset resources are much more common for T1–T3, and to a lesser extent for T4, whereas datasets that naturally support T5 and state-related analysis are fewer and are more often derived from CPS testbeds than from protocol-format benchmarks. PRE evaluation is therefore still most mature at the message and field levels, while process-aware semantic understanding and behavior reconstruction depend on a smaller body of domain-specific datasets.

V. SECURITY, PRIVACY, AND ETHICAL CONSIDERATIONS

As PRE becomes increasingly effective at uncovering the structure and semantics of industrial communication protocols, it also raises important security, privacy, and ethical concerns. In industrial settings, PRE-derived knowledge can facilitate protocol-conformant attacks, enable the inference of sensitive operational information from network traffic, and expose proprietary protocol designs. This section examines these risks and their implications for ICSs.

A. Security Issues

PRE fundamentally lowers the barrier for conducting protocol-conformant attacks against ICSs. By recovering message formats, field semantics, and interaction logic from network traces, PRE enables attackers to move beyond malformed or random traffic and instead generate well-formed messages that closely resemble legitimate control communications [29]. Such capabilities directly undermine defenses that rely on syntactic

deviations or simple traffic anomalies as primary detection signals [22].

As PRE techniques advance toward semantic inference and protocol state machine reconstruction, attackers gain increasingly fine-grained understanding of device behaviors and command sequences. This knowledge facilitates precise manipulation of device states and control workflows, enabling attacks that are both semantically valid and operationally effective [16]. As a result, attacks evolve from protocol-conformant interactions to more sophisticated manipulations that exploit device logic and control dependencies, weakening detection approaches based on control invariants [23].

Furthermore, some PRE approaches rely on active interaction or probing to validate inferred hypotheses [5]. While effective for analysis, these techniques also demonstrate that even protocol-compliant traffic can alter device states or disrupt time-sensitive control logic [76]. In adversarial settings, similar interaction patterns may be exploited to induce unsafe states or degrade system availability, particularly in industrial environments where safety margins are narrow and timing assumptions are critical.

B. Privacy Issues

PRE also raises significant privacy concerns by enabling the extraction of sensitive operational and process-level information from industrial network traffic [87]. Unlike traditional traffic analysis, which often treats payloads as opaque, PRE aims to recover the semantics encoded within protocol fields, including process variables, control parameters, and device states [3]. Such information may reveal detailed insights into industrial processes, production logic, and operational conditions that are not intended to be externally observable.

Through large-scale or long-term traffic analysis, PRE can further expose temporal patterns, control cycles, and correlations among devices, allowing inference of production schedules, process dependencies, and system bottlenecks. In industrial settings, these inferred semantics may constitute proprietary process knowledge or sensitive operational intelligence, the disclosure of which could harm competitiveness or system security [15]. Such privacy risks are amplified when PRE is applied to traffic collected from shared networks, monitoring infrastructures, or Internet-facing industrial services. Even in the absence of explicit identifiers, reconstructed protocol semantics and behavior patterns may enable fine-grained fingerprinting of industrial deployments [1], raising concerns about data ownership, consent, and the unintended secondary use of inferred process information [102].

C. Ethical Issues

The analysis and disclosure of proprietary industrial protocols through PRE introduce distinct ethical considerations. Many industrial communication protocols are custom-designed by vendors and intentionally undocumented, forming part of a manufacturer's intellectual property and competitive advantage [66]. PRE techniques that recover protocol specifications, message semantics, or interaction logic can expose proprietary protocol knowledge without vendor consent.

Beyond intellectual property concerns, public disclosure of private protocol details may increase the exposure of deployed systems to security risks, particularly for legacy devices that cannot be easily patched or upgraded [28]. In practice, such disclosures may reveal previously unknown attack surfaces or

protocol behaviors, while corresponding mitigations remain unavailable. In such cases, PRE research may unintentionally shift risk onto operators who lack the resources or authority to mitigate newly revealed vulnerabilities.

Moreover, the publication of detailed protocol artifacts and semantics raises questions about responsible disclosure and research boundaries. While PRE is often conducted for benign or defensive purposes, the same outputs may be readily repurposed for large-scale reconnaissance, exploitation, or unauthorized system manipulation [27]. Balancing scientific transparency with the potential for misuse remains an open ethical challenge for the PRE research community.

VI. FUTURE RESEARCH DIRECTIONS

Building on the challenges identified across PRE tasks, as well as the associated security, privacy, and ethical issues, we highlight several future research directions that are critical for advancing the robustness, generality, and responsible deployment of PRE techniques in industrial settings.

Data-centric benchmarking and evaluation: Many challenges in PRE arise from the characteristics of traffic traces, including limited diversity, insufficient coverage of protocol behaviors, and varying data volume across studies. These factors substantially affect inference outcomes, yet are rarely modeled or evaluated explicitly. A problem is how to define PRE dataset quality in a task-aware manner, since different tasks may depend on different properties of the underlying traces, such as message diversity, field variability, behavior coverage, or temporal context. Future work should focus on constructing benchmark datasets with controlled characteristics and standardized evaluation protocols, together with metrics that describe dataset properties in addition to algorithmic performance. This would provide a stronger basis for interpreting results and comparing methods.

Dependency-aware joint PRE across tasks: The most existing PRE systems adopt loosely coupled pipelines in which message typing, syntax inference, semantic inference, and state machine reconstruction are treated as independent stages, allowing errors from early stages to propagate to downstream inference. A key future direction is the development of dependency-aware joint PRE frameworks that explicitly model cross-task dependencies and perform unified inference. Rather than committing to a single intermediate result, such frameworks should preserve uncertainty and exploit cross-task consistency signals, including semantic coherence and state-transition constraints, to iteratively refine hypotheses. Moving beyond pipeline-style architectures is essential for robustness under complex and heterogeneous industrial protocols.

Large language model (LLM)-enabled semantic representation and reasoning for PRE: Recent advances in LLMs create new opportunities for PRE by making it possible to reuse semantic knowledge across protocols at a much larger scale [59], [103]. A key challenge is how to use LLMs to support semantic inference without divorcing PRE from trace-based evidence. Future work should investigate trace-conditioned semantic modeling, where recovered fields, value distributions, and interaction contexts are aligned with LLM-generated semantic hypotheses under explicit evidence constraints. This includes retrieval over protocol specifications, documentation, or prior PRE artifacts, with mechanisms for determining whether a semantic interpretation is supported by observed traces rather than merely plausible in language space. Addressing this problem could improve

field-role inference and cross-protocol semantic transfer while reducing hallucinated interpretations in industrial PRE.

Security-aware and ethically grounded PRE methodologies: Increasingly capable and automated PRE techniques raise the risk of misuse in industrial settings where protocol semantics are tightly coupled with physical processes. Future research should embed security, privacy, and ethical considerations directly into the design of PRE methodologies, rather than treating them as external constraints. This includes developing disclosure-aware evaluation practices, limiting the exposure of sensitive operational semantics in released artifacts, and exploring abstractions that preserve analytical value while reducing exploitability. Establishing community norms for responsible dataset sharing and artifact release will be critical to ensuring that PRE advances defensive understanding without disproportionately increasing real-world risks. In addition, future work should explore practical applications of PRE in industrial environments, such as protocol-aware intrusion detection and asset discovery, to ensure that methodological advances translate into deployable defensive capabilities.

VII. CONCLUSION

This article presented a task-oriented review of network-trace-based PRE for industrial protocols. Despite recent advances that have significantly expanded the scope and capability of PRE techniques, fundamental challenges remain in data dependency, inference reliability, and evaluation practice. Moreover, the growing effectiveness of PRE raises important security, privacy, and ethical considerations in industrial contexts. Addressing these issues is essential for advancing PRE from isolated case studies toward more systematic and deployable methodologies. The insights from this paper provide a structured basis for future PRE research and support the effective and responsible use of PRE techniques for ICSs.

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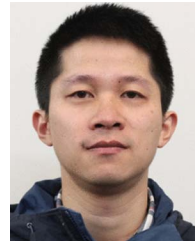
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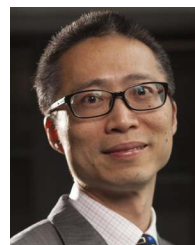
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